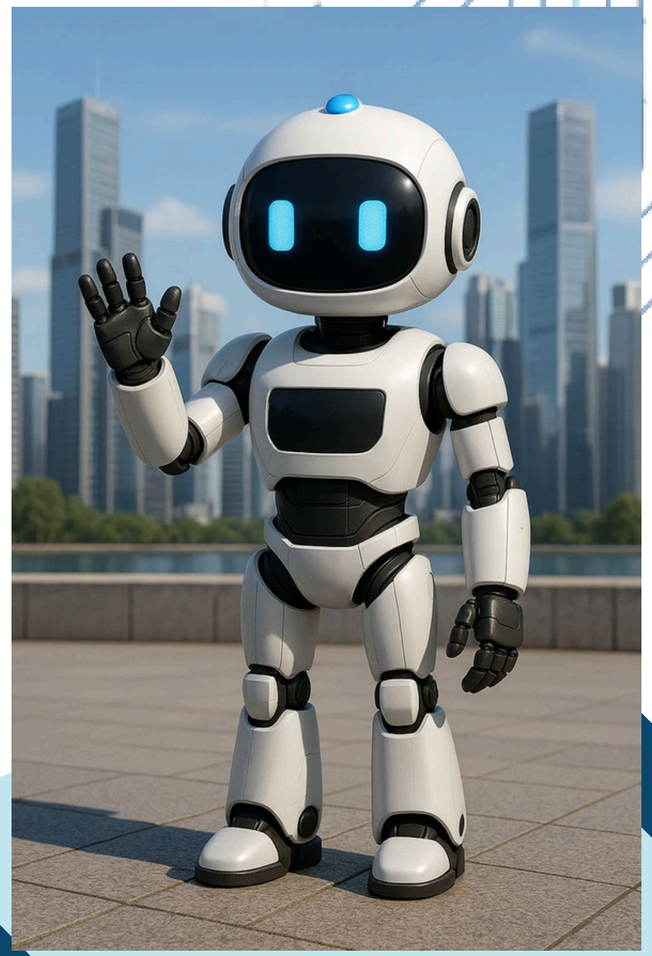
Agentic AI

Welcome to a journey into Agentic AI

January 11 , 2026



prepared by: Daniyah Aleid

TABLE OF CONTENTS

- Foundations
- Architecture
- Design patterns
- Frameworks
- Evaluation
- Guardrails
- Practice & Future





EVOLUTION FROM TRADITIONAL AI

- Single-purpose, **reactive**
- Requires explicit invocation
- Input → Process → Output
- No persistence

Traditional AI

Generative AI

- Conversational, **general-purpose**
- Still reactive and stateless
- Prompt → Generate → Response
- **Turn-based** interaction

Agentic AI

- **Autonomous**, goal-directed
- **Proactive**, stateful, **tool-using**
- Goal → Plan → Act → Reflect
- Continuous operation

Agentic AI Definition

AI systems that can **autonomously** perceive their environment, **make decisions**, **plan** sequences of **actions**, and **execute** them to **achieve** specified **goals** with minimal human intervention.





Use Cases

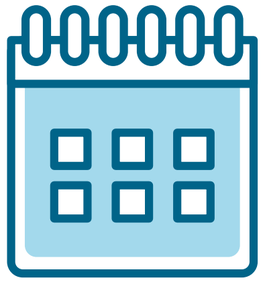
Research Assistant



Customer Service



Appointment Scheduling



Flight Booking



Invoice Processing





Key Properties

- **Autonomy**: Self-directed behavior
- **Goal-oriented**: Works toward objectives
- **Parallelization**: Handles multiple tasks simultaneously
- **Proactive**: Takes initiative
- **Adaptive**: Learns from experience

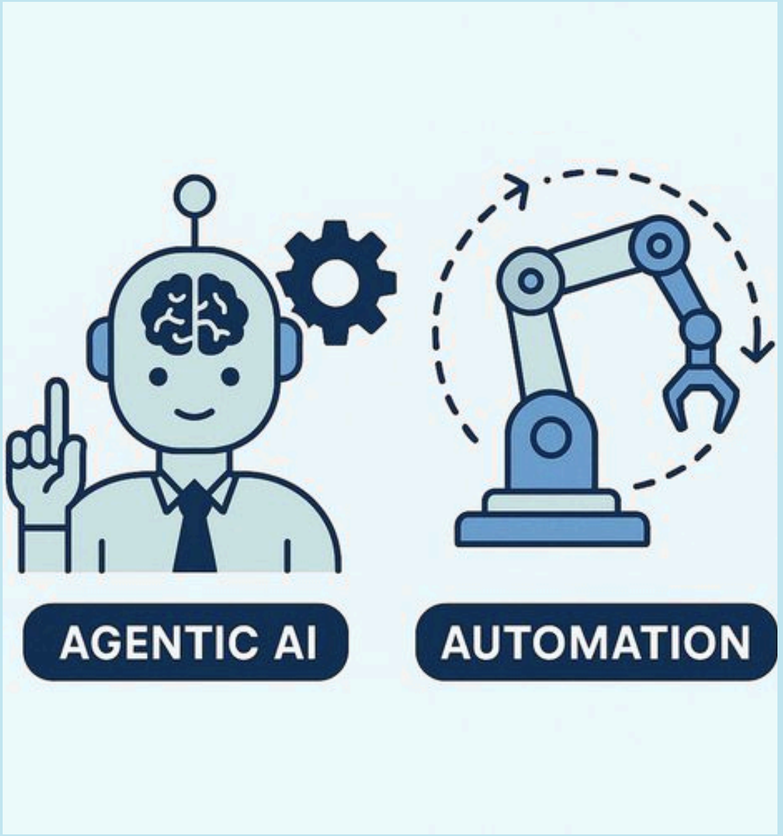
Core Capabilities

- **Perception**: Understand context
- **Planning**: Decompose goals
- **Reasoning**: Logical inference
- **Action**: Execute in environment
- **Learning**: Improve over time



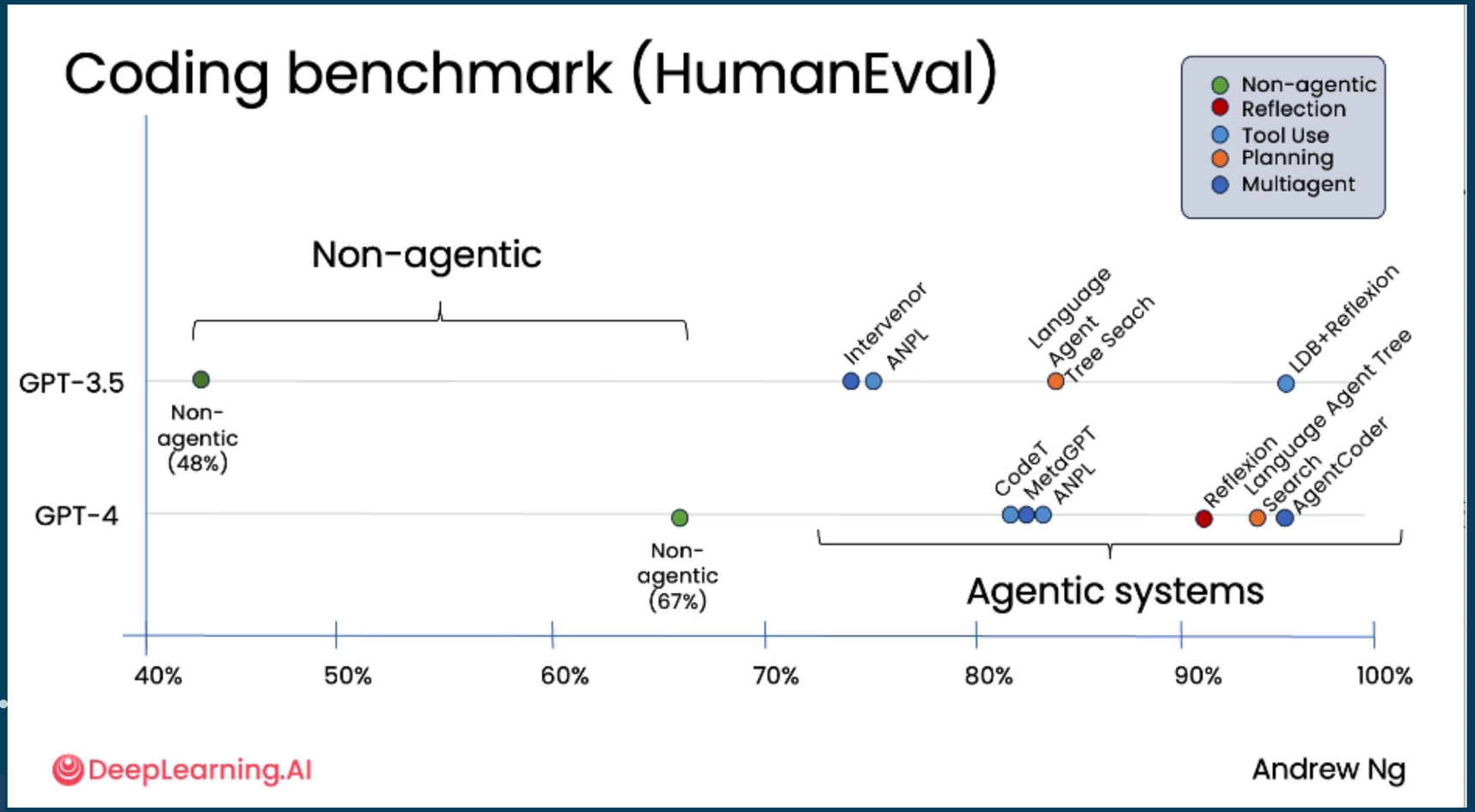
Agentic AI vs. Automation

Automation executes pre-defined tasks reliably, while **Agentic AI** autonomously reasons, adapts, and handles complex workflows.



builtin.com

why agentic AI is better?



DeepLearning.AI

Agentic systems achieve higher success rates in coding tasks compared to non-agentic models.

SYSTEM ARCHITECTURE

Reasoning Engine (The Brain)



- LLM-based core.
- reasoning
- decision-making.

Memory (The Knowledge)



- Short-term
- Long-term

Planning Module (The Strategist)



- Breaks down goals into actionable steps.

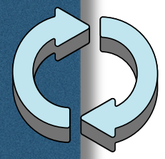
SYSTEM ARCHITECTURE

Tools
(The Hands)



- Connects to external APIs and tools.
- executes function calls.

Reflection
(The Critic)

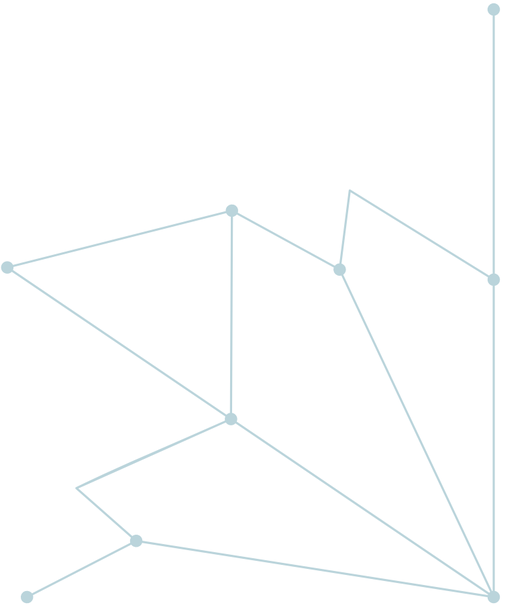


- Evaluates performance.
- Identifies errors.

Agentic Design Patterns

Common design patterns includes:

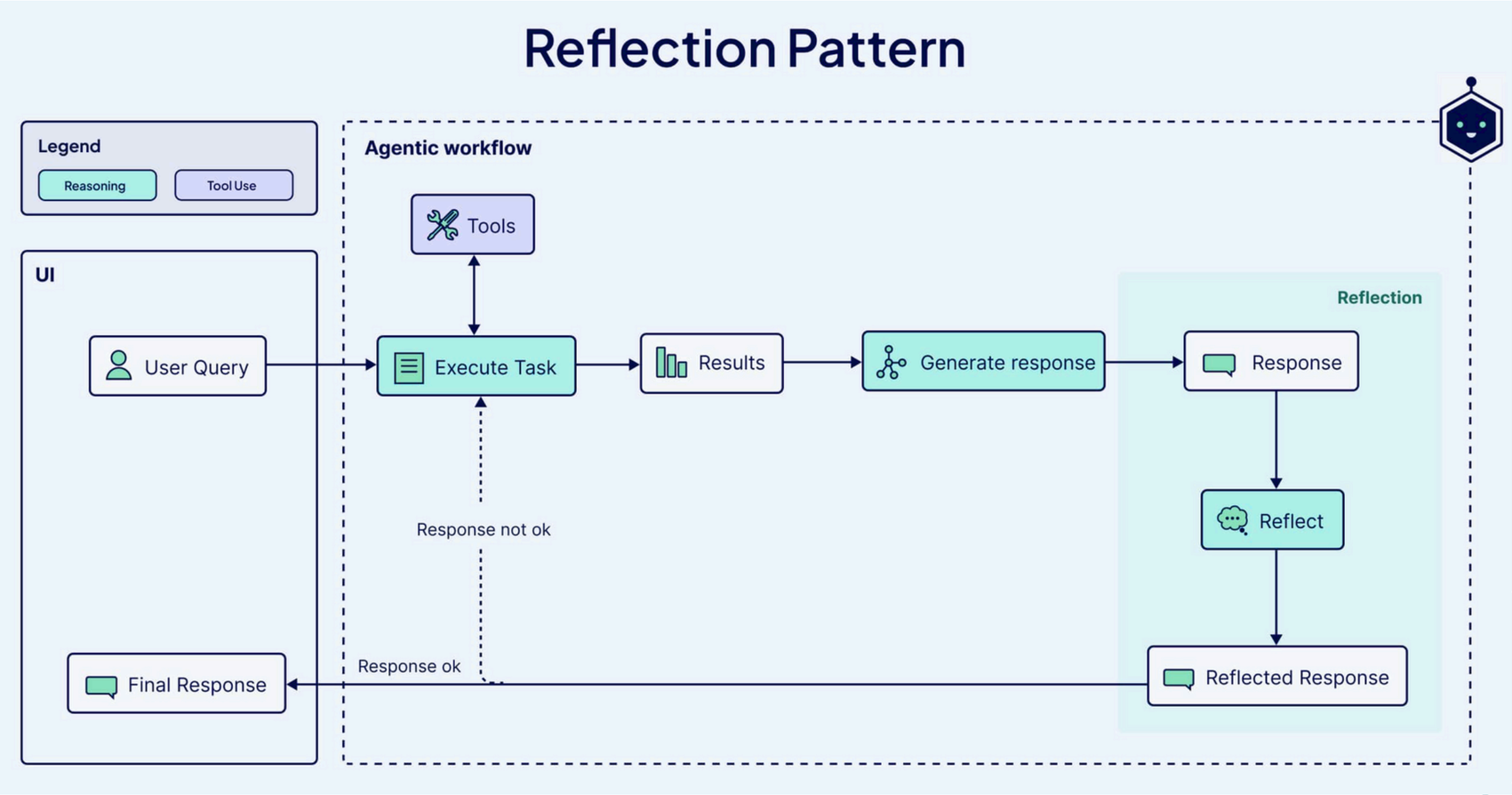
1. Reflection
2. Tool Use
3. Planning
4. Multi-Agent Pattern



1. Reflection

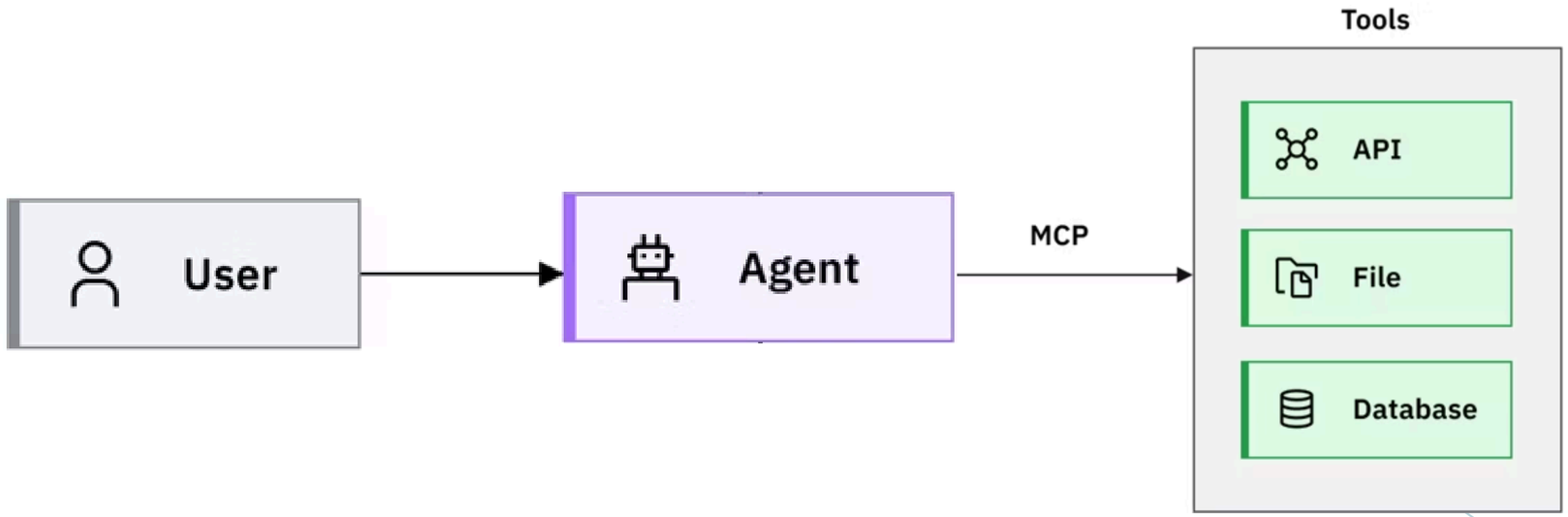
- AI **reviews** its own outputs
- Detects **errors** and inconsistencies
- **Self-corrects** and **improves iteratively**

1. Reflection



2. Tool Use

- Use **external tools**, APIs, or software to achieve tasks.
- Enables **action beyond text generation**.



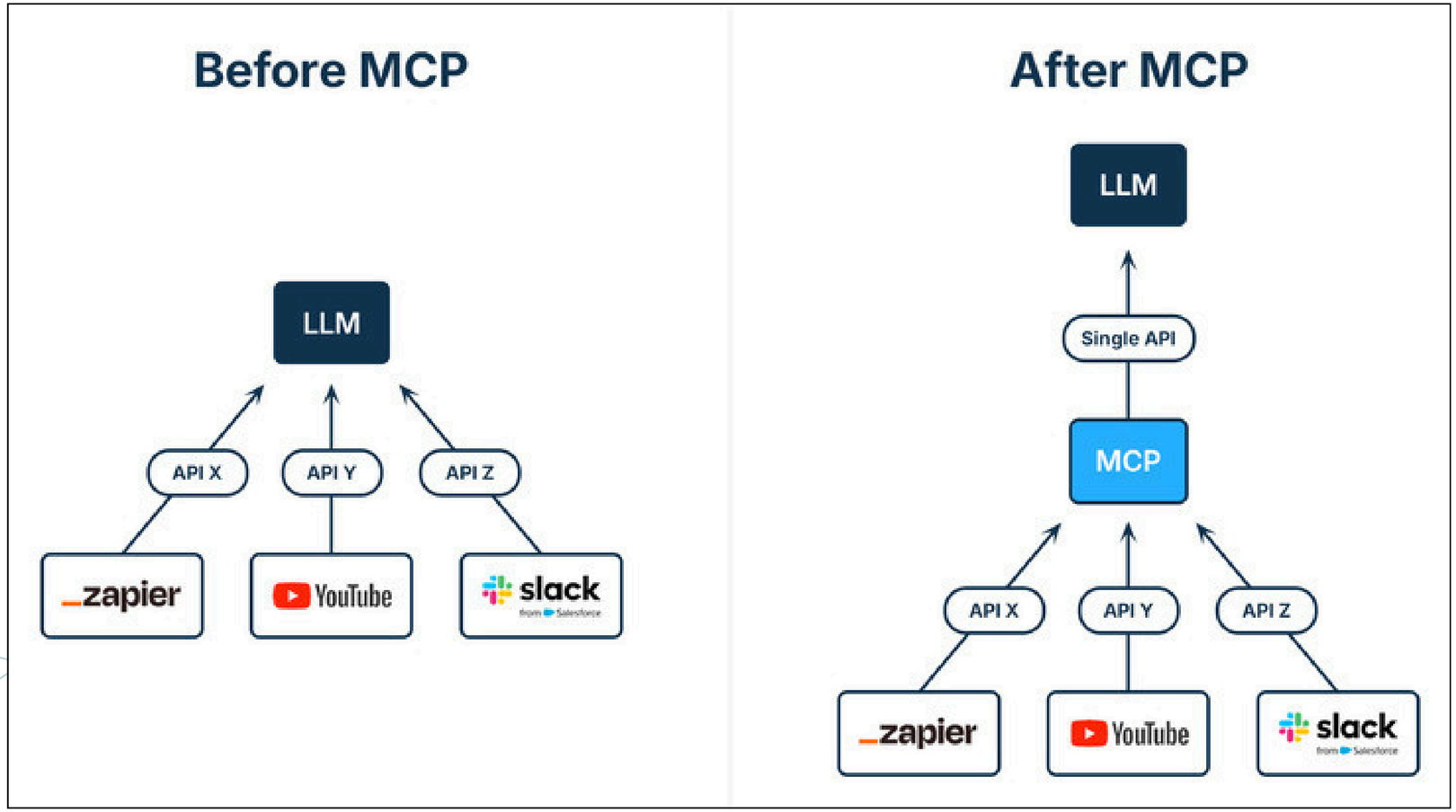
IBM.com

Model Context Protocol (MCP)

- open-source standard for **connecting** AI applications to external systems.
- Enable AI applications to connect to **data sources**, **tools** and workflows, to **access** key information and **perform tasks.**

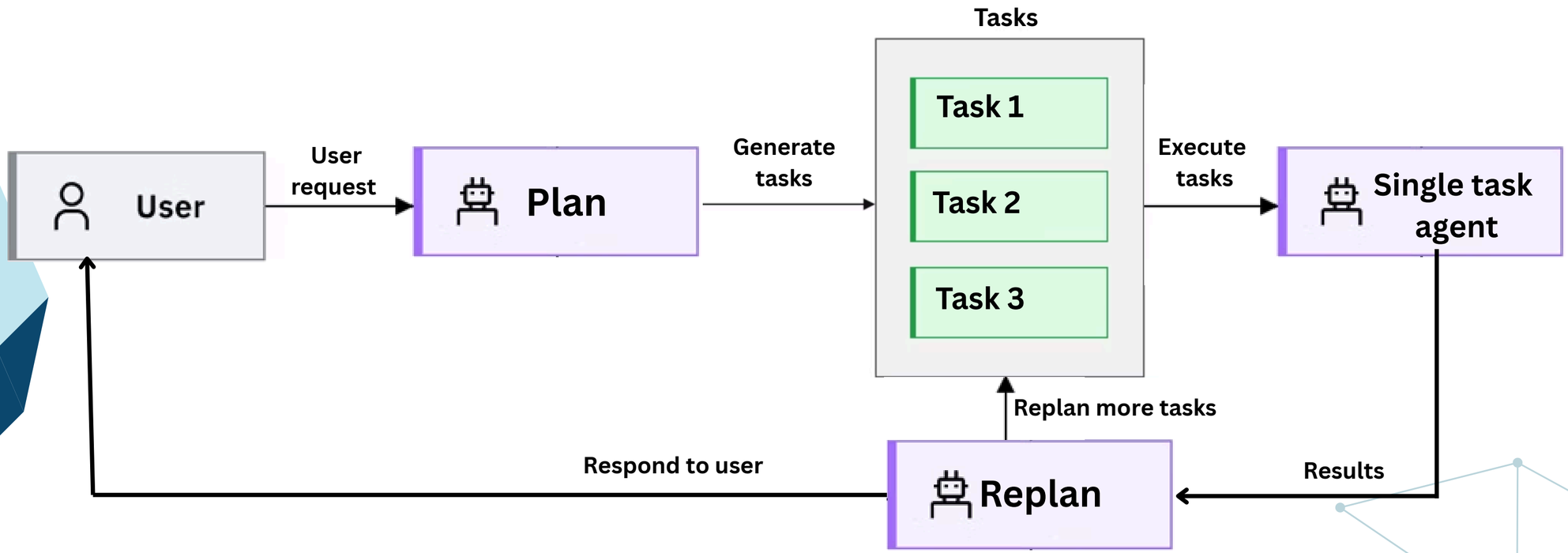
modelcontextprotocol.io

Model Context Protocol (MCP)



civo.com

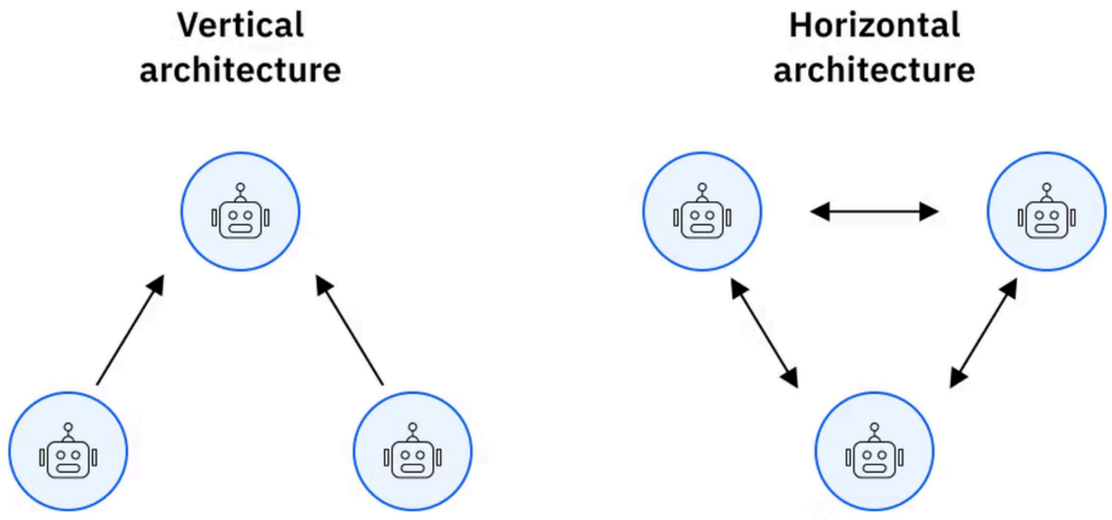
3. Planning



blog.langchain.com

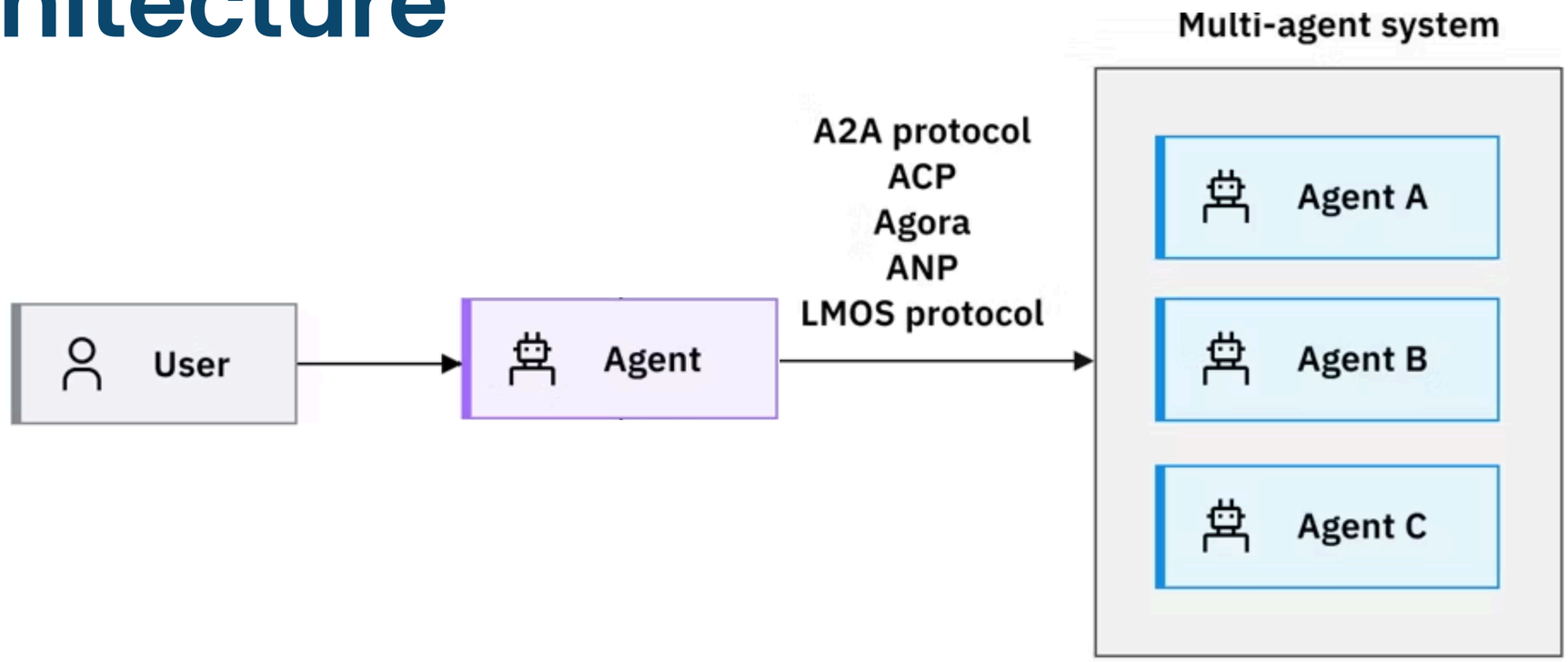
4. Multi agent architecture

In multi-agent architectures Each agent **specializes** in a specific domain while seamlessly **collaborating** to solve complex problems.



IBM.com

Communication in Multi agent architecture



Agentic AI frameworks

With code:









google ADK

OpenAI ADK

No code:







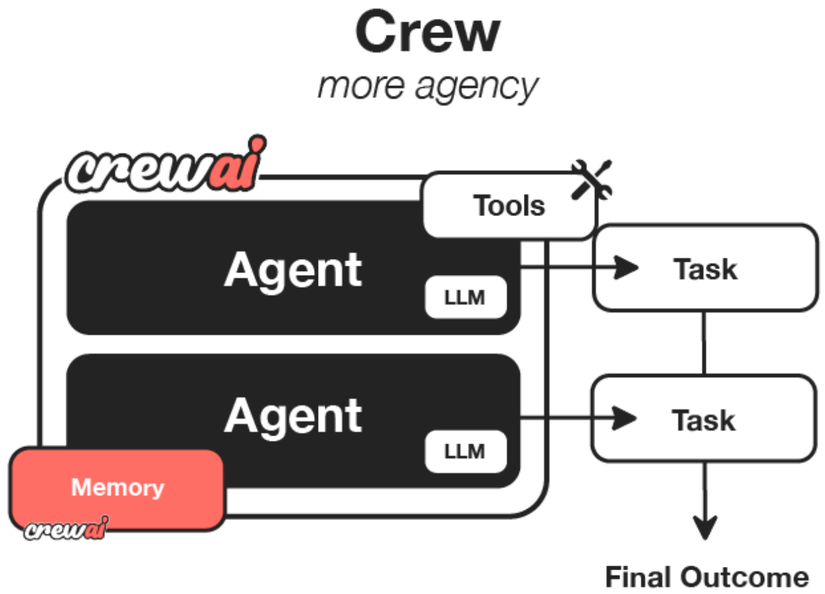




Codebasics

CrewAI

CrewAI is an open-source framework for building and coordinating multiple AI agents. It supports both **code-based** development and a **visual drag-and-drop** interface, allowing you to create complex agent workflows with minimal coding.



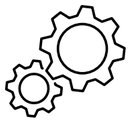
docs.crewai.com

What you can do with CrewAI (Example)

Research Automation (Multi-Agent Research Team)



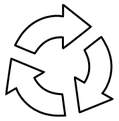
- **Research Agent**: Searches the web for reliable sources



- **Analysis Agent**: Extracts insights & compares findings

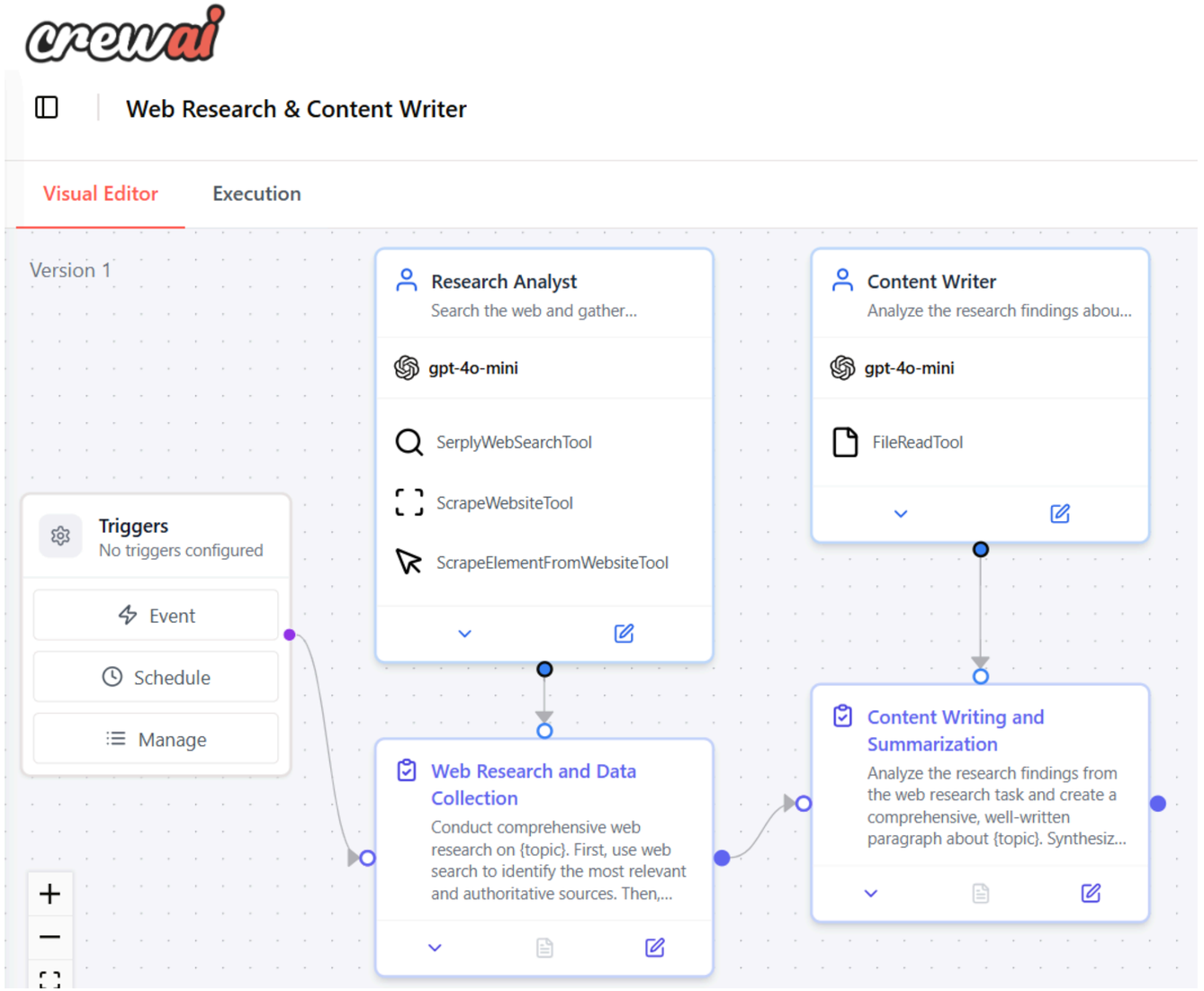


- **Writer Agent**: Creates a structured, professional summary



- **Reviewer Agent**: Checks clarity, accuracy, and tone

CrewAI

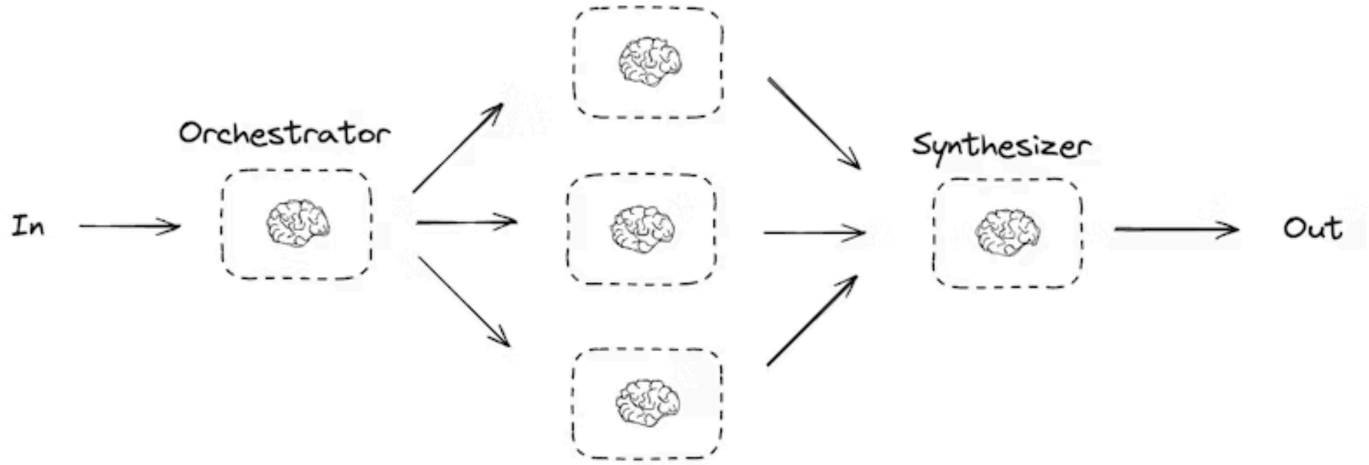


crewai

docs.crewai.com

LangGraph

LangGraph LangGraph uses a **graph-based** approach to define and execute agent workflows, making it suitable for complex, dynamic systems.
Agents are defined as **nodes, and edges** represent the connections between them, determining the order in which agents execute.



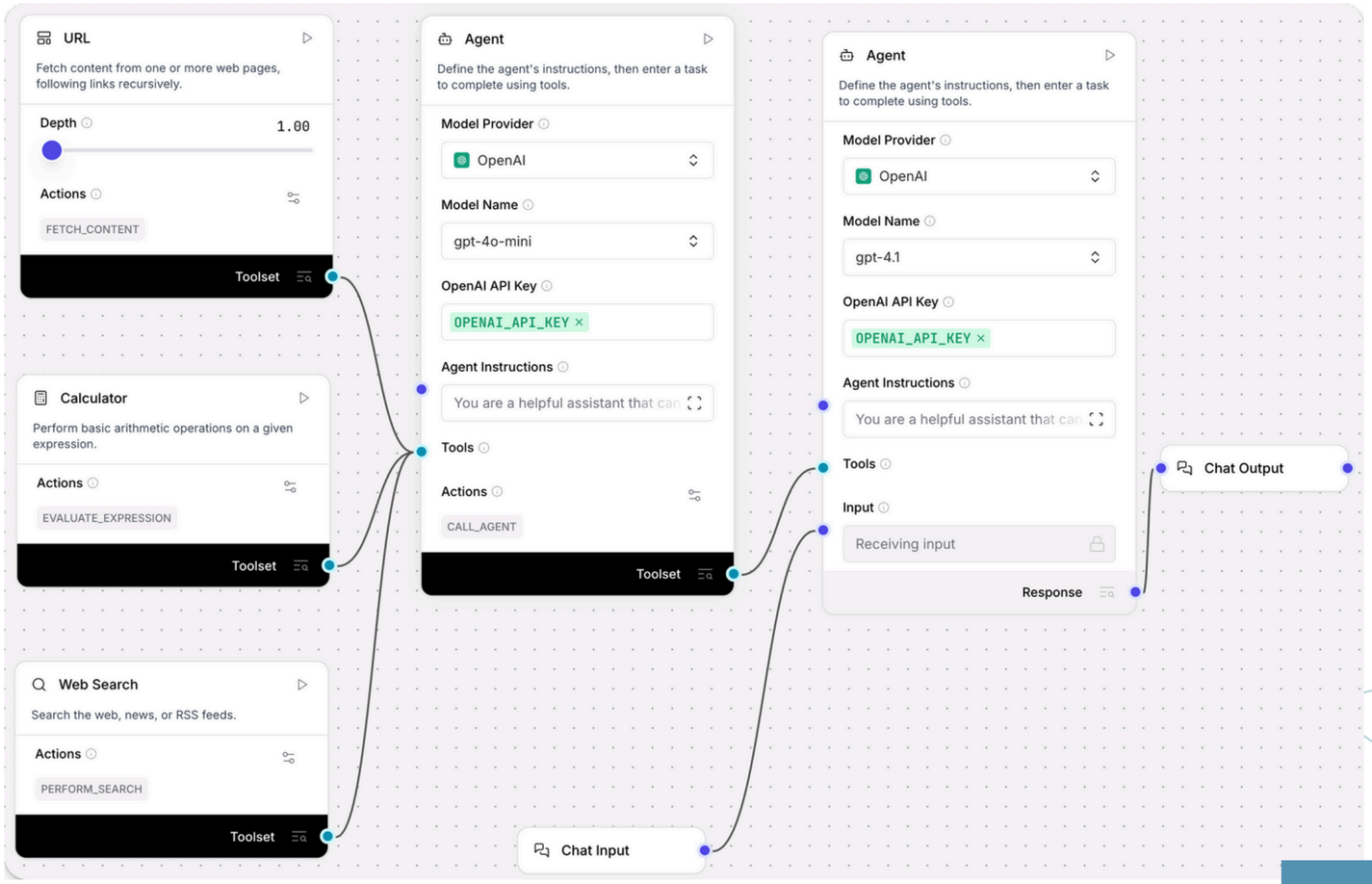
docs.langchain.com

LangFlow

LangFlow is a tool for building and deploying **AI agents** and **MCP servers**. It includes built-in components and supports major LLMs, vector databases, and a growing library of AI tools.

docs.langchain.com

LangFlow

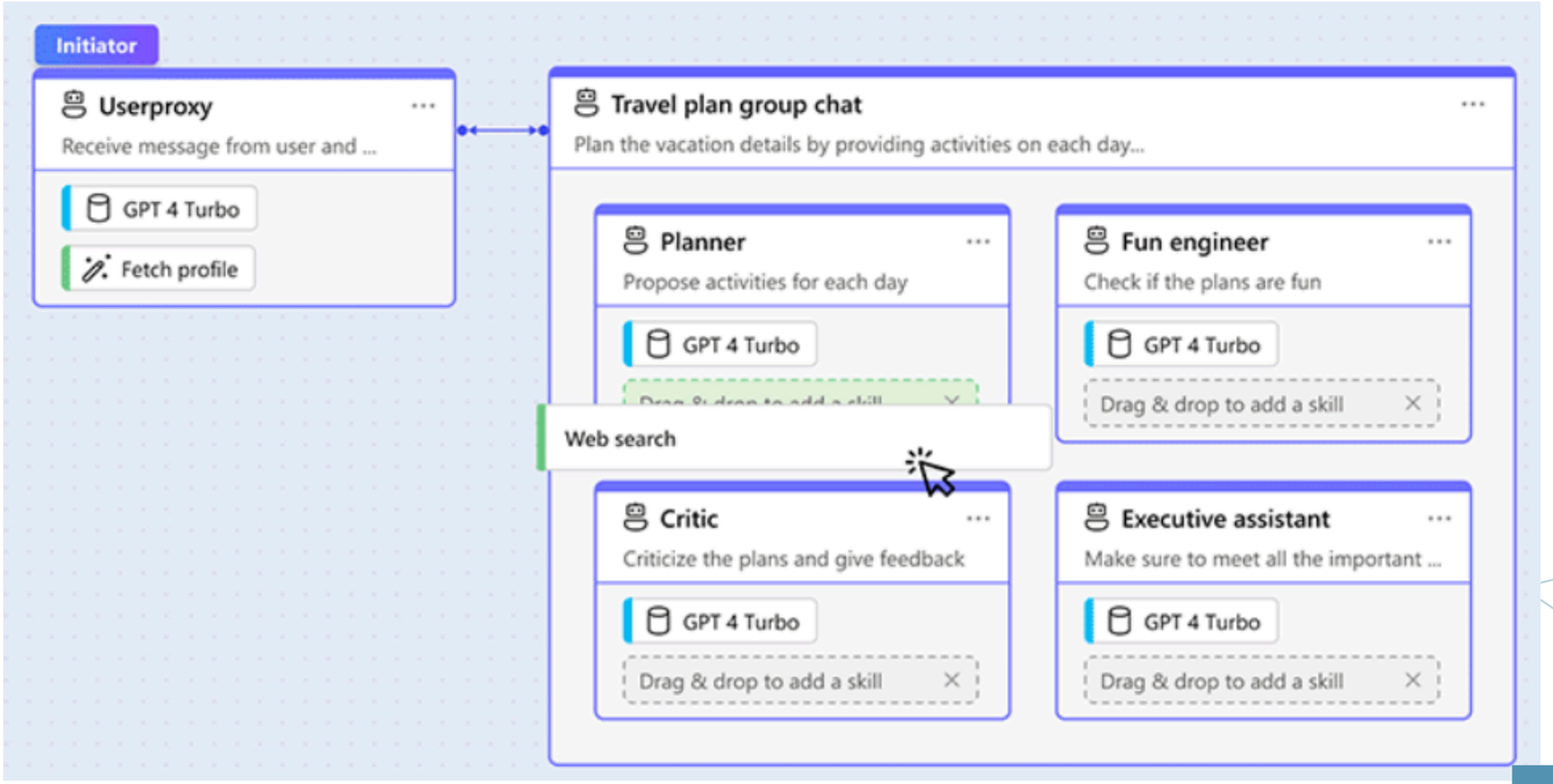


docs.langchain.com

AutoGen

AutoGen Studio

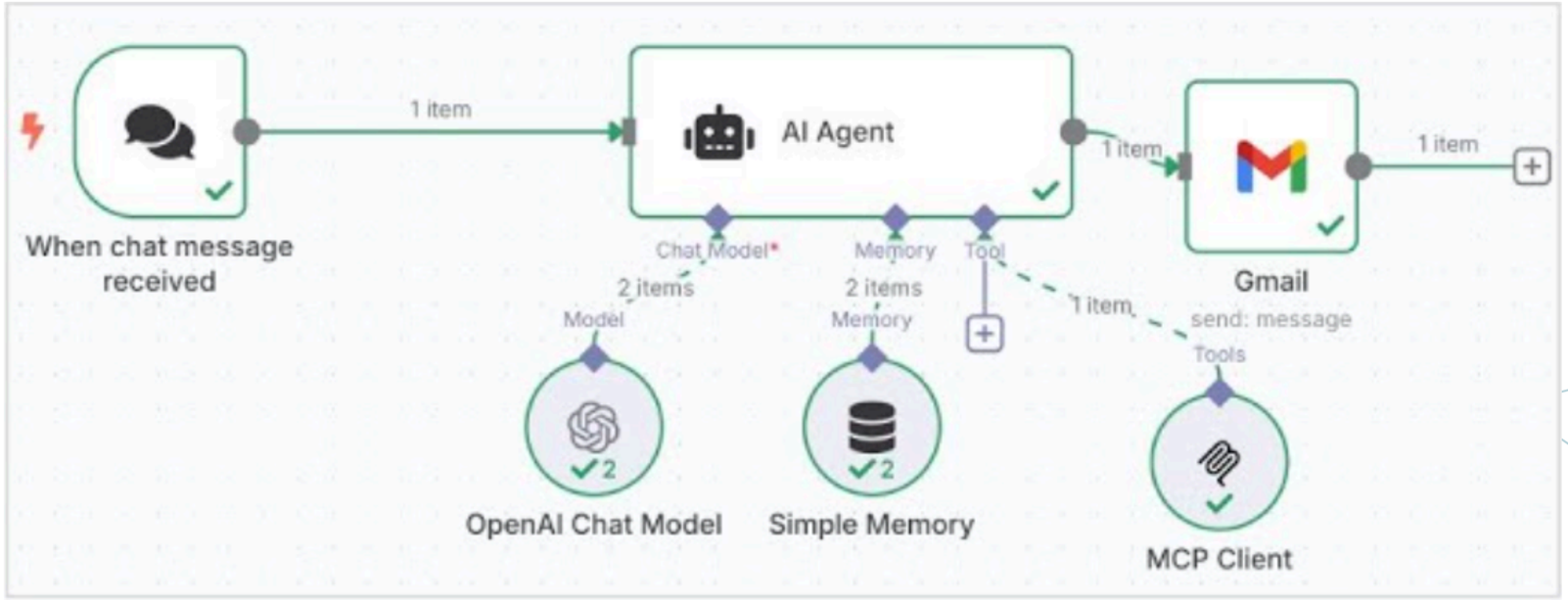
AutoGen is a multi-agent conversation framework designed to orchestrate autonomous, event-driven systems.



microsoft.com

n8n

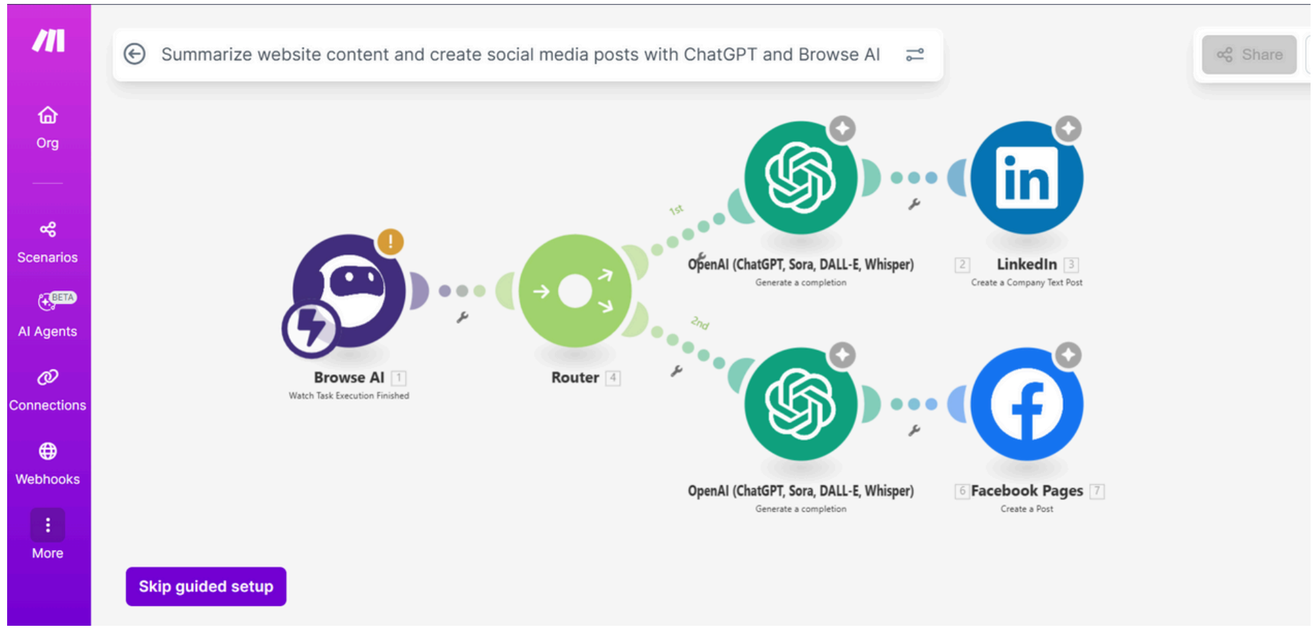
n8n is an open-source workflow tool that connects AI models with tools, APIs, and data sources to build autonomous, multi-step Agentic AI systems **without coding.**



n8n.com

make

make is a no-code workflow platform that connects AI models with apps and APIs, enabling automated, multi-step workflows and agent-like behavior.



make.com

Manus as a leading example of Agentic AI

- **manus** is a powerful agentic AI system that goes beyond simple chat. It acts autonomously, breaks down tasks, uses tools, and completes multi-step workflows with minimal guidance.
- It integrates **memory, reasoning, and tool usage,** making it capable of analyzing documents, generating reports, and automating enterprise tasks.



manus.im

Manus Capabilities and Toolset

- **Web Browsing & Research**: Manus can navigate the internet, bypass simple obstacles, and synthesize information from multiple sources to provide up-to-date research.
- **Code Execution & Development**: Manus can build software, automate data processing, and fix bugs in real-time.
- **Data Analysis & Visualization**: It can ingest large datasets, perform complex statistical modeling, and generate professional charts and reports.

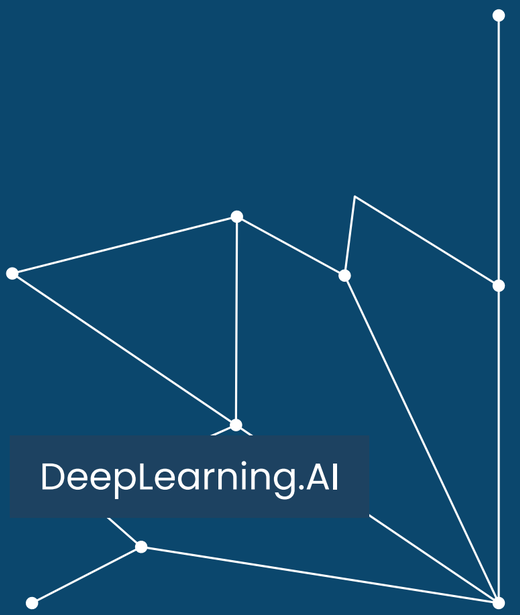
Agentic AI Evals

Objective evaluation

- Evaluation based on measurable, quantifiable metrics with no human interpretation needed.
- Ex. Error rates, execution speed.

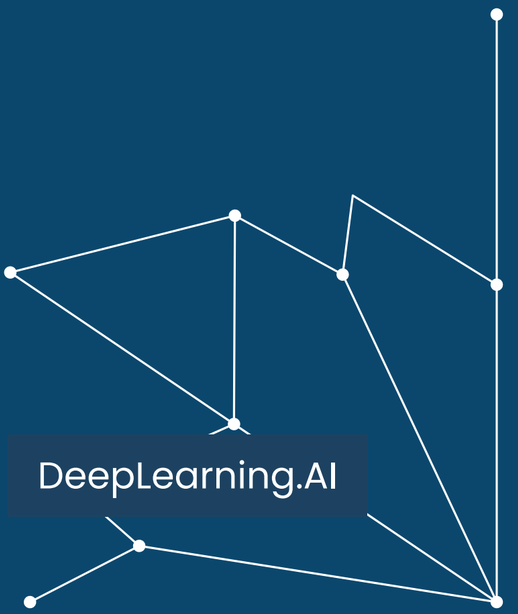
Subjective evaluation

- Evaluation based on human judgment of agent performance, behavior, and quality.
- Ex. Helpfulness, politeness, and tone.



Agentic AI Evals

Objective evaluation	Subjective evaluation
Metrics-based	Metrics-based
Repeatable	Variable
Quantitative	Qualitative
Measures correctness	Measures quality
No human bias	Human perception involved



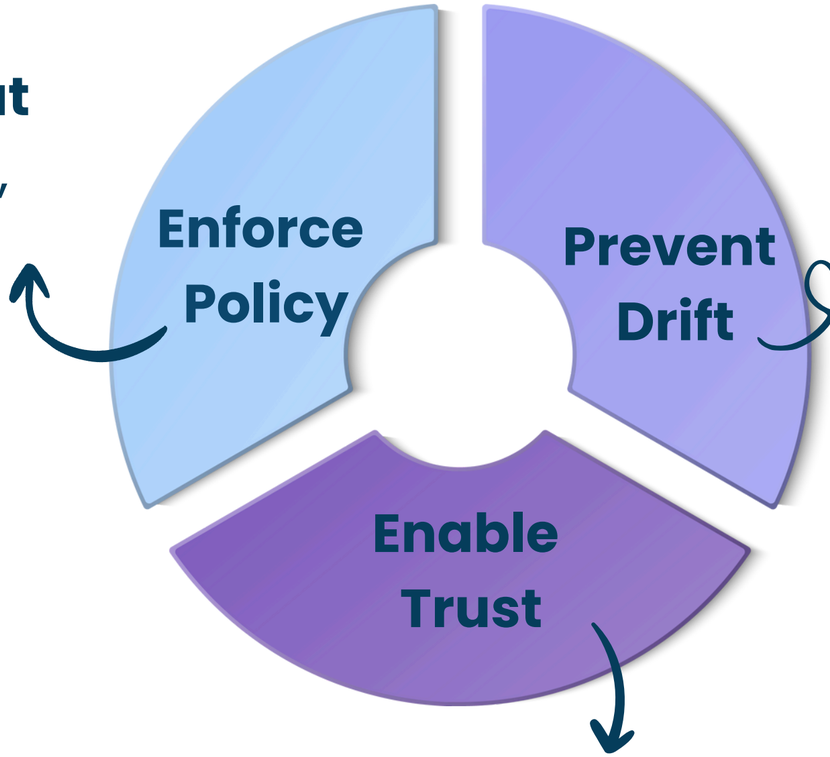
Why Guardrails Matter for Agentic AI?

Autonomous agents operate at speed and scale, but **unchecked actions** can create security, compliance, and operational **risk.**



Guardrails:

Define and **restrict what agents can do**; by role, context, or risk level.



Ensure agents **do not deviate from business intent** or compliance frameworks.

Make agent decisions **transparent, explainable**, and **auditable**.

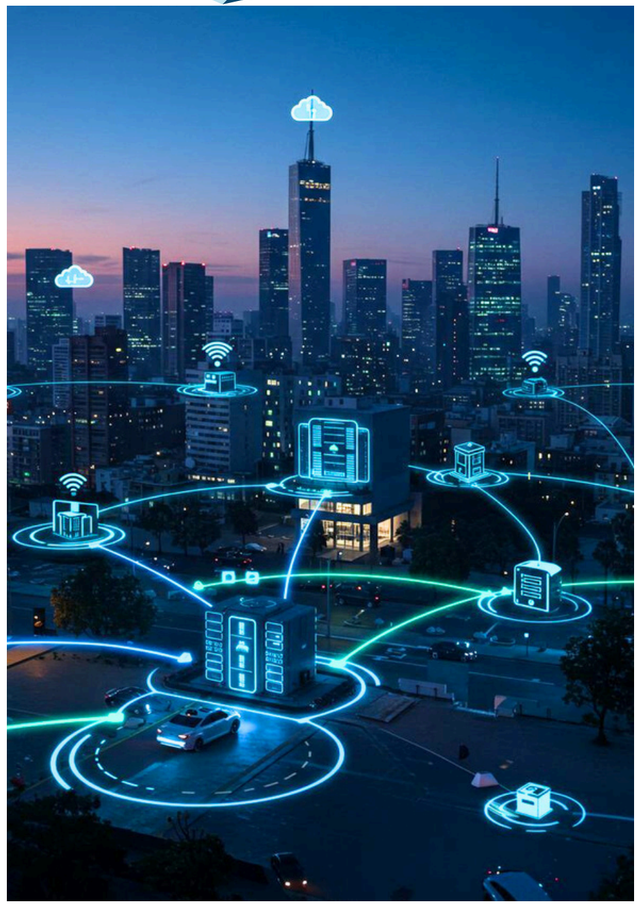
How to Take Advantage of Agentic AI?

Manufacturing

- Predictive maintenance agents
- Supply chain optimization
- Quality control automation
- Real-time monitoring & alert systems



How to Take Advantage of Agentic AI?



Telecommunication

- Autonomous network monitoring & optimization
  Ex.Real-time KPI analysis (latency, throughput, packet loss, call drop rate)
- Automated troubleshooting and customer support
  Ex.Root cause identification using historical failure patterns
- Predictive maintenance for infrastructure
  Ex.Health monitoring of network elements (base stations, radios, power units)

How to Take Advantage of Agentic AI?

Healthcare

- Personalized treatment recommendations
- Medical Imaging Agent
- Clinical Decision Support (CDSS)



Thank you

Prepared by: Daniyah AlEid

Supervised by: Dr. Abdulaziz AlQabbany



Artificial Intelligence and Robotics Institute
King Abdulaziz City for Science and Technology